MILITARY SPECIFICATION

BLOWER, ANTI-EXPOSURE SUIT VENTILATION, GENERAL SPECIFICATION FOR

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers anti-exposure suit ventilation blowers, operable from 115/200 volt, 400 cycle AC aircraft power systems.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawing, and publication, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-D-70327	Drawings, Engineering and Associated
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-M-3171	Magnesium Alloy; Processes for Corrosion Protection of
MIL-P-116	Preservation, Methods of
MIL-P-5633	Packaging and Packing of Aircraft Material in Steel Shipping Containers
MIL-P-6064	Packing of Lightweight Aircraft Accessories
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-129	Marking for Shipment and Storage

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DRAWINGS

Air Force-Navy Aeronautical Standard Drawing

AND10398	Metals - Definition of Dissimilar
MS22053(WEP)	Blower, 400 CPS, Continuous Duty, Class A, Garment Ventilation, Mark-5 Anti-Exposure Suit
MS33543(ASG)	Criteria-Temperature and Altitude Range Self Cooled Electric Equipment

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards, Use of

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The blowers furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 General.- Whenever the requirements of this specification and the detail specification or drawing conflict, the requirements of the detail specification or drawing shall govern. Any deviation from this specification or from subsidiary specifications or drawings, where applicable, must be specifically approved in writing by the procuring activity.

3.3 Materials.- Materials used in the manufacture of units shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specifications will not constitute waiver of Government inspection.

3.3.1 Metals.- Metals shall be of a corrosion-resistant type or suitably processed to resist corrosion.

3.3.2 Dissimilar metals.- Unless otherwise suitably protected, dissimilar metals, such as brass, copper, or steel, shall not be used in intimate contact with magnesium, aluminum or their alloys. When protection is used, it shall be of such type that a low-impedance path is offered to radio-frequency currents. Dissimilar metals are defined in Drawing AND10398.

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3.3.3 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.3.3.1 Standard parts.- Standard parts (MS, AN or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Any standard part, of the same part number, shall be readily interchangeable as applied within the unit. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Design and construction.-

3.4.1 Detail specifications and drawings.- Units shall conform to the requirements on the applicable drawing or detail specification. Applicable detail specification or drawings are listed on Standard MS22053(WEP).

3.4.2 Operating Conditions.- The unit shall be designed to furnish cooling air continuously under the following conditions or natural combination thereof in any position when the input voltage and frequency are varied over the ranges permitted by the detail specification or drawing.

3.4.2.1 Pressure Altitude.- Barometric pressures ranging from 30 inches Hg to the barometric pressure at the maximum altitude indicated by the detail specification or drawing. The pressure may remain constant, or may vary at a rate as high as 1.5 inch Hg/sec.

3.4.2.2 Ambient temperature.- Ambient temperature shall be defined on the applicable drawing or detail specification for any particular unit.

3.4.2.3 Humidity.- Relative humidity ranging up to 100 percent, including conditions wherein condensation will take place on the equipment.

3.4.2.4 Sand and dust resistance.- Under conditions of airborne sand and dust particles.

3.4.2.5 Salt spray.- Atmosphere containing salt-laden moisture.

3.4.2.6 Fungus.- When exposed to fungus growth in accordance with Procedure I of specification MIL-E-5272.

3.4.2.7 Operating position.- When installed in any position.

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3.4.2.8 Overspeed.- The unit shall be capable of operating at no load at a frequency of 500 cps as required by section 4.5.9.

3.4.2.9 Shock.- When subjected to a shock condition as required by the Shock tests, Procedure II, of Specification MIL-E-5272.

3.4.2.10 Input power.- The input power shall be in accordance with MIL-E-7894. The type power used shall be as specified in the detail specification, drawing, or both.

3.4.2.11 Duty cycle.- The duty cycle shall be continuous operation.

3.4.3 Detailed requirements.- The detailed requirements shall be as specified in this blower specification, drawing, or both.

3.4.4 Simplification.- Simplicity of design resulting from use of the same part for as many applications as possible in a given unit and in units of different ratings, is highly desirable.

3.4.5 Lubrication.- The lubrication shall be such that it will insure satisfactory operation. Units shall not require lubrication during 1,000 hours of operation.

3.4.6 Electrical connections.- Internal wiring of the unit shall be in accordance with the applicable drawing or detail specification and shall terminate as shown thereon. If input and output contactors are used, they shall be deenergized simultaneously.

3.4.7 Rotor.- The units shall be so constructed that the rotor may be readily removed without special tools.

3.4.8 Voltage and frequency.-

3.4.8.1 Voltage.- The rated input voltage shall be 115 volts line to neutral. Unless otherwise stated in this specification all voltages shall be assumed to be on a line to neutral basis. Satisfactory operation at rated load and frequency shall be obtained with the input voltage variations specified in the applicable drawing.

3.4.8.2 Frequency.- The rated frequency shall be 400 cycles per second. Motors shall operate satisfactorily at rated load and voltage over the frequency range of Type I power, MIL-E-7894 unless otherwise specified in the applicable motor drawing, detail specification, or both.

3.4.8.3 Combined voltage and frequency.- Blowers shall operate satisfactorily under any combination of voltages and frequencies as defined in paragraphs 3.4.8.1 and 3.4.8.2.

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3.4.9 Brush type ac motors.- When brushes are used additional requirements applicable to brush type AC motors will be made effective.

3.4.10 Protection.- All ventilating openings shall be of such size as not to permit passage of a 5/16-inch-diameter rod. No area on the unit with all covers in place shall be exceptionally vulnerable to damage.

3.4.11 Life.- The unit shall be so designed that when operating under any temperature or altitude conditions indicated by the detail specification or drawing, the useful life of the unit shall be at least 1,000 hours. The brushes (when used) and any readily accessible plug-in type component may be replaced after 500 hours of operation.

3.4.12 Rotation.- The direction of rotation of the motor rotor or fan blade shall be as specified in the applicable blower drawing.

3.4.13 Thermal protection.- When thermal protection is a requirement, one of the following types of protection shall be provided as required by the applicable blower drawing:

a. To prevent smoke hazard.

b. To provide optimum operational life consistent with maximum power capability under all operable conditions.

c. To provide maximum emergency blower output without being a fire hazard.

3.4.14 Oil proof.- When specified by the applicable blower drawing, the blower shall be designed and constructed to operate under conditions of oil leakage upon the motor as defined in paragraph

3.4.15 Explosion proof.- When specified by the applicable blower drawing, the motor shall be explosion-proof as defined in paragraph

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of specification MIL-D-5028.

3.6 Screw threads.- Screw threads shall conform to the requirements of Specification MIL-S-7742.

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3.7 Safety wiring and staking.- Accidental loosening of screws and screw parts and other connections shall be prevented by safety wiring (0.032-inch minimum OD), where practicable, staking, or other approved method. Washers and cotter pins, where used, shall be assembled in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.8 Finish.-

3.8.1 Aluminum or aluminum-alloy parts.- Where practicable, aluminum-alloy parts shall be treated for corrosion resistance in accordance with specification MIL-A-8625 or specification MIL-C-5541, or approved equivalent. The film deposited by this treatment shall be removed from the actual contact area of all surfaces required to act as a path for electrical current and from the local areas under screws, nuts, or the like used for assembly or mounting purposes, to provide an adequate bonding connection.

3.8.2 Magnesium-alloy parts.- Wherever practicable, magnesium-alloy parts shall be surface treated in accordance with Specification MIL-M-3171 to provide protection against corrosion.

3.8.3 Electrical sheet steel assemblies.- Electrical sheet steel assemblies, such as armatures, field poles, etc., shall be processed by acceptable commercial methods, to provide protection against corrosion.

3.8.4 Plating.- Cadmium plating shall be in accordance with Specification QQ-P-416, Type I or II, as applicable and of a class that is adequate to achieve the degree of protection required.

3.9 Performance.- The blower shall satisfy the performance requirements specified in Section 4 when subjected to the following tests:

Heating, speed
Dielectric Strength
Performance at sea level
Performance at altitude
Temperature rise
Acoustical noise
Power factor
Overspeed
Operating position
Starting current
Two-phase operation
Input current balance
Acceleration
Shock
Vibration
Life
Humidity
Salt Spray
Sand and Dust
Fungus

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3.10 Markings.-

3.10.1 Terminal Markings.- Marking of terminals shall be in accordance with Standard MIL-STD-195 unless otherwise shown by the applicable motor drawing, detail specification, or both. When deviations from MIL STD-195 exist, a diagram or chart of the terminals shall be affixed to the motor.

3.11 Identification of product.- Parts and sub-assemblies of blowers shall be marked with the part number and the manufacturer's name or trademark where practicable.

3.11.1 Nameplate.- Each blower shall be marked for identification in accordance with Specification MIL-M-7911 and shall contain the following information:

AC Aircraft Motor
For use on 115/200 Volt, 400 Cycle System

HP _____
RPM _____
Amps/Phase _____
Phases _____
Power Factor _____
Duty _____(Continuous)

MIL Part No. (or Detail Specification No.)
Stock No. _____
Manufacturer's Part No. (or Identification) _____
Manufacturer's Serial No. _____
Contract or Order No. _____
Manufacturer's Name or Trademark _____

3.11.1.1 Restriction on size of trademark.- The manufacturer's name or trademark shall be not larger than other letters appearing on the nameplate.

3.11.2 Use of ANor MIL designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.12 Workmanship.- All machine surfaces shall have a smooth finish, and all details of manufacture, including the preparation of parts and accessories, shall be in accordance with the best practices for high-quality electrical equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, lacquering, riveting, clearance between soldered connections and ruggedness.

3.13 The airflow shall be measured with the equipment shown in figure 1.

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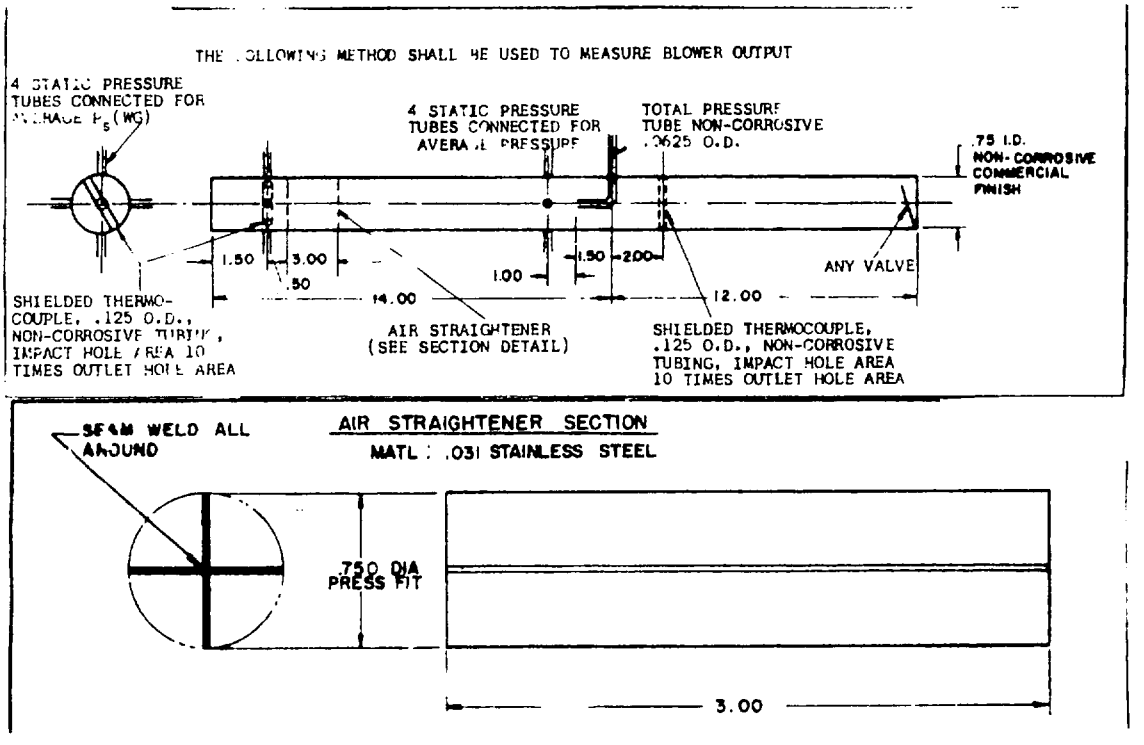


FIGURE 1

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Tests

UNIT NO. 1

(a) Examination of product
(b) Heating, speed
(c) Dielectric strength
(d) Performance at sea level
(e) Performance at altitude
(f) Temperature rise
(g) Acoustical noise
(h) Power factor
(i) Overspeed
(j) Two-phase operation
(k) Starting current
(l) Input current balance
(m) Shock

UNIT NO. 2

(a) Examination of product
(b) Heating, speed
(c) Dielectric strength
(d) Performance at sea level
(e) Performance at altitude
(f) Temperature rise
(g) Acoustical noise
(h) Power factor
(i) Overspeed
(j) Two-phase operation
(k) Starting current
(l) Input current balance
(m) Acceleration
(n) Humidity

UNIT NO. 3

(a) Examination of product
(b) Heating, speed
(c) Dielectric strength
(d) Performance at sea level
(e) Temperature rise
(f) Power factor
(g) Overspeed
(h) Two-phase operation
(i) Vibration
(j) Fungus
(k) Salt spray

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UNIT NO. 4

(a) Examination of product
(b) Heating, speed
(c) Dielectric strength
(d) Performance at sea level
(e) Temperature rise
(f) Power factor
(g) Overspeed
(h) Two-phase operation
(i) Life
(j) Sand and dust

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3.13.1 The formula shown below shall be used to calculate the airflow:

$$CFM = 1097\ A\sqrt{\frac{h_t - h_s}{D}}$$

$$D = \frac{1.325\ P}{T}$$

WHERE:

A = Pipe Area (Sq. Ft.)
h_t = Total Pressure (Inches of Water)
h_s = Average Static Pressure (Inches of Water)
P = Absolute Air Pressure In Tube (Inches of Mercury)
T = 460 + T_1 (°F)
T_1 = Air Temperature In Tube Near Total Pressure Tube

4. QUALITY ASSURANCE PROVISIONS

Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1 Classification of tests.- The inspection and testing of a unit shall be classified as follows:

(1) Qualification Tests: Qualification tests are those tests performed on samples submitted for approval as qualified products.
(2) Acceptance Tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 Qualification tests.-

4.2.1 Sampling instructions.- The qualification test samples shall consist of 8 units upon which qualification is desired. The manufacturer shall furnish all samples and shall be responsible for accomplishing the required tests on 4 samples. Upon completion of the tests, the units tested, the 4 untested units, together with test reports, shall be identified as required and forwarded to the activity responsible for qualification as designated in the letter of authorization from that activity (see section 6.5). The test reports shall include a statement that the samples submitted have been subjected to and have complied with the dielectric strength test (section 4.5.3). Each unit shall be accompanied by a set of spare brushes if brushes are used. Units shall be accompanied by two copies of outline and detail assembly drawings thereof and by two copies of sample instructions with illustrations and diagram, if necessary, covering the installation of the units.

4.2.1.1 At the option of the procuring activity, any or all tests may be repeated. If the manufacturer desires to submit new units for test purposes, units which have passed tests at the manufacturer's plant shall accompany these items.

4.2.2 Tests.- The Qualification tests shall consist of all the tests of this specification, as described under "Test Methods". The tests shall be conducted on each of four units as shown below and must be conducted essentially in the order listed for each unit.

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4.2.2.1 Disassembly and inspection.- At the conclusion of the Qualification tests, the units will be disassembled and inspected for excessive wear and defects.

4.2.3 Rejection and retest of qualification samples.- Units which have been rejected or returned to the manufacturer for any reason during Qualification tests, may be reworked or have parts replaced to correct defects. Before resubmitting the unit, full particulars concerning the rejection and the corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the test activity and to the procuring activity. Tests shall not be resumed until such a report is received. Where Qualification tests are conducted under the auspices of the manufacturer, the procuring activity shall be advised upon failure of qualification sample and of the action taken by the manufacturer with regard to failure.

4.3 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.3.1 Individual tests.- Each blower shall be subjected to the following tests described under "Test Methods":

(a) Examination of Product
(b) Performance at sea level
(c) Line Current Balance (Polyphase Only)
(d) Dielectric Strength
(e) Thermal Protection - Short test only (When thermally protected)

In addition, the motor shall be subject to any other tests specified herein which the Inspector, with the approval of the Procuring Agency, considers necessary to determine conformance with the requirements of this specification.

4.3.2 Sampling tests.- Sampling tests, if required, shall be as specified by the procuring activity.

4.3.3 Rejection and retest.- Blowers which have been rejected may be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.4 Test conditions.- Unless otherwise specified, each test in this section shall be made under the following conditions.

4.4.1 Ambient.- Tests for which no ambient temperature is specified shall be performed at an ambient temperature of 25° ± 15°C. If different maximum ambient temperatures are indicated for the control component and the power component of the unit by the detail specification or drawing, the components shall be separated for those tests requiring maximum ambient temperatures and each subjected simultaneously to its respective maximum ambient temperature. Separated components shall be operated as a unit by means of a connecting cable.

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4.4.2 Altitude.- Tests will be run at approximately sea-level altitude.

4.4.3 Minimum performance.- The air output from the blower shall meet or exceed the values specified by the applicable drawing.

4.4.4 Voltage measurement.- The input and output voltage shall be measured at the terminals of the unit.

4.4.5 Input voltage.- Nominal input voltage as shown on applicable drawings shall be used.

4.4.6 Frequency.- The frequency of the input voltage shall be 400 cycles per second.

4.4.7 Position.- The unit shall be mounted with its major axis horizontal and its base down.

4.5 Test methods.-

4.5.1 Examination of product.-

4.5.1.1 Qualification tests.- Each unit subassembly and part shall be examined as the Inspector may deem necessary to determine compliance with this specification and the applicable drawing or detail specification with respect to material and workmanship, standard parts, simplification, lubrication, electrical connections, rotor removal, brush accessibility and position, protection, threaded parts, finish, interchangeability, marking, instructions, dimensions, and weight.

4.5.1.2 Individual tests.- Each unit subassembly and part shall be examined as the Inspector may deem necessary to determine compliance with this specification and the applicable drawing or detail specification with respect to material and workmanship, lubrication, electrical connections, threaded parts, finish, interchangeability, marking instructions, dimensions, and weight.

4.5.2 Heating, speed.- Provision shall be made for determining frequency, speed, input voltage and current, input watts, and output airflow. While the motor is cold the resistance and temperature of the field shall be determined for use in calculating the final field temperature rise after operation at rated load. A temperature measurement may be made by thermocouple on the frame or field structure in order to determine when the temperature has leveled off prior to taking a final resistance reading on the field winding. The blower shall be considered to have reached its continuous operating condition when the rate of rise of field temperature above the then exisitng ambient is no more than $1^{\circ}C$ ($2^{\circ}F$) for a period of 5 minutes.

4.5.2.1 Heating.- The ability of the blower to deliver rated output airflow under the following conditions for continuous or rated duty cycle operation shall be demonstrated. The final temperature of the blower shall not be cause for rejection provided the blower meets all other requirements of the specification.

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(a) Rated frequency, rated voltage.
(b) Minimum frequency, maximum steady state voltage.
(c) Maximum frequency, minimum steady state voltage.
(d) For polyphase motors only, minimum frequency, minimum steady state voltage on one phase, maximum steady state voltage on all other phases.

Unless otherwise stated in the applicable blower drawing, the frequency limits of Type I power, MIL-E-7894 shall be assumed. The minimum and maximum steady state voltages shall be 102 and 124 volts respectively.

At the option of the Procuring Service, a life test run under one of the above conditions of voltage and frequency in lieu of the rated voltage and frequency may be performed in addition to the above tests.

4.5.2.2 Speed.- At the completion of the heat run at rated input voltage and frequency, the blower speed shall be within 5 percent of the value specified by the applicable blower drawing.

4.5.3 Dielectric strength.-

4.5.3.1 Test conditions.- The dielectric strength of the various components shall be made by subjection to the following test voltages at commercial frequency. These tests may be conducted on the components prior to final assembly. Units failing to meet the following tests shall be rejected:

(a) Circuits which operate at a voltage of less than 50 volts and below shall be tested at 500 volts (rms) for 1 minute or 600 volts (rms) for 1 second.

(b) Circuits which operate at a voltage exceeding 50 volts shall be tested at 1,250 volts (rms) for 1 minute or 1,500 volts (rms) for 1 second.

4.5.3.2 Capacitors.- Capacitors shall be subjected to and shall withstand for 1 minute a test voltage of twice the maximum peak voltage encountered during normal operating conditions. Units failing to meet this test shall be rejected.

4.5.4 Performance at sea level.- The air output of the blower shall be measured with the equipment shown in figure 1. During this test the unit shall be operated at or near sea level pressure. For each voltage and frequency condition the amount of output air shall not be less than the minimum value at the required static pressure shown on the applicable drawing.

4.5.4.1 Room temperature operation.- ($25^{\circ}C \pm 15^{\circ}C$)

(a) At minumum input voltage and minimum frequency, the static pressure shall be adjusted to the minimum value required by the applicable drawing. The amount of air flow shall be calculated using the formula shown in 3.13.1.

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(b) At rated input voltage and minimum frequency, the preceding test shall be repeated.

(c) At maximum input voltage and minimum frequency, the preceding test shall be repeated.

(d) Repeat tests (a) to (c) above at rated input frequency.

(e) Repeat tests (a) to (c) at maximum input frequency.

4.5.4.2 Extreme temperature operation (-55°C).- Repeat 4.5.4.1 at -55°C.

4.5.4.3 Extreme temperature operation (+71°C).- Repeat 4.5.4.1 at +71°C.

4.5.5 Performance at altitude.- For this test, rated input voltage and frequency shall be used. The blower shall be instrumented so that bearing, winding, stator and other hot spot or critical temperature readings can be checked and recorded. The blower shall be operated under the following altitude and temperature conditions required by the applicable drawing for altitude:

(a) One hour at standard test conditions.

(b) One hour at sea level and maximum temperature.

(c) One hour at 20,000 feet altitude and maximum temperature.

(d) One hour at sea level and minimum temperature.

(e) One hour at 20,000 feet altitude and minimum temperature.

(f) One hour test to demonstrate continuous operation at

the temperature altitude conditions within the range established by curve II of Drawing MS33543 from 20,000 to 40,000 feet.

4.5.6 Temperature rise.- During the course of tests under 4.5.4 and 4.5.5 the temperature increase of the air due to compression and energy dissipation through the blower shall not be more than the maximum allowed by the applicable drawing.

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4.5.7 Acoustical noise.- The audible noise level shall be measured with a General Radio sound-level meter type 1551-B or equivalent meter. When measured within an area one foot from the unit, the acoustical noise shall not be more than the maximum specified on the applicable drawing.

4.5.8 Power factor.- At rated voltage and frequency, the speed, input watts and input current shall be determined. The power factor shall be not less than that shown by the applicable drawing when a minimum is specified.

4.5.9 Overspeed.- The blower shall be capable of withstanding for a short duration, an overspeed condition that would result from imposing a frequency of 500 cps, without causing disintegration of the blower or the motor.

4.5.10 Operating position.- During the course of other tests, it shall be ascertained that the operation of the motor in any position does not adversely affect power output, heating, lubrication, and other operating characteristics.

4.5.11 Starting current.- The starting line current shall be measured using a recording oscillograph while starting the motor at no load when at room ambient temperature; this line current shall not exceed 700 percent of the full load line current or 250 amperes, whichever is smaller. The line current shall return and stay within 300 percent of full load line current within 0.2 second after first application of input phase voltage. The current oscillogram shall indicate a definite current peak and shall have no tendency to remain at peak value.

4.5.12 Two-phase operation.- The blower shall be capable of continuously delivering 50% of the specified output with any one of the input power leads disconnected.

4.5.13 Input current balance.- For polyphase motors the maximum difference in line currents shall not be greater than 10 percent of the minimum steady state current when measured at rated load and balanced terminal voltage.

4.5.14 Acceleration.- The blower shall be mounted on a centrifuge. The centrifuge shall then be operated at a speed that will produce an acceleration of ten gravitational units; the blower shall then be rotated 180 degrees about an axis perpendicular to the axis of the centrifuge arm and the test repeated. The blower shall be mounted on the centrifuge in four additional positions such that the accelerating force is applied in both directions through each of the remaining axes of the blower. For each position of the blower, the centrifuge shall be operated at a speed that will produce an acceleration of 10 gravitational units. The acceleration of 10 gravitational units in each test shall be stabilized and maintained for a period of not less than 1 minute.

4.5.15 Shock - The unit shall be subjected to the Shock tests of Specification MIL-E-5272, Procedure II, Equipment Operation. Damage or failure of any part of the unit during this test shall be cause for rejection. Following these tests, the unit shall pass the test specified in 4.5.4.1 (d) of this specification. The unit shall then be subjected to the Shock tests of Specification MIL-E-5272, Procedure II, Equipment Crash-Safety.

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4.5.16 Vibration.- The unit shall be subjected to the Vibration Tests, Procedure XII of Specification MIL-E-5272, with the following exceptions:

(a) During the Durability test, Section 4.7.10.1 of Specification MIL-E-5272, the unit shall be operated at full load. Following this test, the unit shall pass the test of section 4.5.4.1(2).

(b) During the Performance test, section 4.7.10.2 of Specification MIL-E-5272, the frequency of vibration range shall be 20 to 500 cycles. The vibration excursion shall take 12 to 15 minutes with no one frequency held longer than 3 minutes. Following this test, the unit shall pass the test of section 4.5.4.1 (d).

Any mechanical damage occurring as a result of the vibration test shall be cause for rejection.

4.5.17 Life.- The blower shall operate for 1,000 hours at rated voltage and rated frequency, in an ambient temperature of 25° ± 15° C. The output air shall be restricted sufficiently to produce forward of the restriction, the required static pressure shown on the applicable drawing. The unit shall be brought to a complete stop and started within 3 minutes at least once every 25 hours. No maintenance of any kind shall be allowed during this test.

4.5.18 Humidity.- The unit shall be subjected to the Humidity tests, Procedure I, of Specification MIL-E-5272. Immediately following this test, the unit shall pass the test specified in 4.5.4.1 (d).

4.5.19 Salt Spray.- The unit shall be subjected to 25 hours of salt spray in accordance with the Salt spray tests of Specification MIL-E-5272. Following this test, the unit shall be washed, and dried for 15 to 20 hours, and then shall meet the requirements of 4.5.4.1 (d). There shall be no evidence of excessive corrosion.

4.5.20 Sand and dust.- The unit shall be subjected to the Sand and dust tests, Procedure I, of Specification MIL-E-5272 at sand and dust velocities of 200 ± 100 fpm. The 8-hour test cycle shall be composed of 30-minute operative and 90-minute inoperative periods. The total operative test duration shall be 2 hours and the inoperative test shall be 6 hours. At the end of the test, the unit shall be removed from the test chamber and subjected to the test specified in 4.5.4.1 (d).

4.5.21 Fungus.- The unit shall be subjected to the Fungus resistance tests, Procedure I, of Specification MIL-E-5272. Immediately following this test, the unit shall meet the test requirements of 4.5.4.1 (d).

4.6 Material test specimens.- Test specimens shall be furnished when requested by the Government for check tests at a Government laboratory to determine conformance with the applicable specification.

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5. PREPARATION FOR DELIVERY.

5.1 Application.- The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation, packaging, and packing.- The units shall be preserved and unit packaged in accordance with Specification MIL-P-116, Method II, using containers conforming to Specification MIL-D-6054.

5.3 Marking of shipments.- Shipping containers shall be marked in accordance with Standard MIL-STD-129. The identification shall be composed of the following information listed in the order shown:

Federal Stock No. or other identification number as specified in the purchase document*
BLOWER, AIRCRAFT, GENERAL SPECIFICATION FOR
MS Part No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Date of manufacture
Name of contractor
Name of manufacturer (if different from contractor)

*NOTE: The contractor shall enter the Federal Stock No. specified in the purchase document or as furnished by the procuring activity. When the Federal Stock No. is not provided or available from the procuring activity, leave space therefor and enter the Stock No. or other identification as provided by the procuring activity.

6. NOTES

6.1 Intended use.- This specification covers the requirements for blowers for use with aircraft 115/200 volt, 400 cycle, a-c systems, to ventilate anti-exposure suits.

6.2 Ordering data.- Requisitions, contracts, and orders should state the part number or detail specification, Sampling tests (4.3.2), and whether domestic or overseas packing is desired (see 5.2). The contractor should pack with each unit assembly one printed copy of simple instructions with illustrations and any necessary diagrams covering the installation of the unit. Prior to printing, two copies should be furnished the procuring activity for approval. Instructions should be printed on paper 8-1/2 by 11 inches or 17 by 11 inches.

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6.3 Qualified Products List.- For procurement purposes, the provisions of this specification will not become effective until publication of the Qualified Products List, to be numbered QPL _____

6.4 All a-c voltages as used in this specification should be considered to be root mean square (rms) values.

6.5 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Naval Weapons. Requests for information pertaining to qualification of products covered by this specification should be addressed to Chief, Bureau of Naval Weapons, RAAE-5, Navy Department, Washington, D. C., with a copy to the Commander, Aeronautical Systems Division, Wright-Patterson Air Force Base, Ohio.

6.5.1 In the case of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the activity responsible for qualification considers sufficient to warrant additional tests.

6.5.2 Upon notification of qualification approval, the manufacturer should provide two reproducible copies of outline and detail assembly drawings. Any changes in the reproducible copies from those submitted with the qualification samples should be indicated in detail.

> Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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